Shipbuilding

Time Limit

2min

Using Grinders

Watch the video

Job Category	Shipbuilding	Subject	Using grinders

I Requirements

– Understand how to combine / disassemble and use the grinder.

II Diagram & Requirements

1) Diagram



Job Category	Shipbuilding	Subject	Using grinders

2) Requirements

① Attach the grinder handle.

② Attach the guard cover to the grinder and use a hex wrench to tighten the screw completely.

③ Insert the flange(만조), disc, and clamping nut in that order.

④ Press the spindle lock on the grinder to lock the rotating shaft.

⑤ Turn on the grinder, perform a usage simulation, then turn off the grinder.

⑥ After the supervisor's confirmation, if instructed, press the spindle lock and disassemble in the reverse order of assembly.

Ⅲ Equipment & Materials

Equipment & Materials	Spec.	Unit	QTY	Remakrs
 Grinder Set	· 4inch Grinder · Handle · Guard Cover · Disc · Clamping Nut · Grinder Spanner · Hex Wrench	SET	1	For 1 person

Time Limit

7min

Pipe Volting

Watch the video

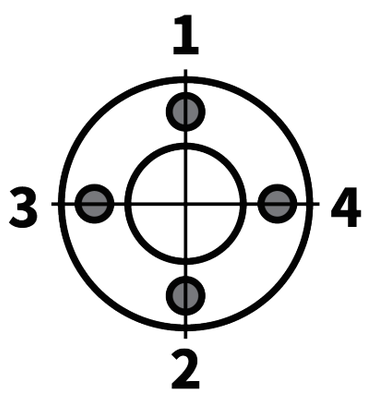
Job Category	Shipbuilding	Subject	Pipe Volting

I Requirements

- Use the tool to join the piping flanges.

II Diagram & Requirements

1) Diagram

Assemble Sequence	Example
	
Result	**Result (Side)**
	

Job Category	Shipbuilding	Subject	Pipe Volting

2) Requirements

① Assemble the flange using the provided materials and tools, and obtain confirmation from the supervising officer.

<order of assembly>

1. Insert the gasket between the flanges and align it.
2. Lightly tighten the bolts, then check the offset between the two flanges.
3. Using a spanner, tighten the bolts and nuts in a diagonal sequence.
4. During the final tightening, confirm that there is no offset on the flange to complete the task.

② After the supervisor's confirmation, disassemble the assembled flange and gasket within the time limit according toinstructions.

Job Category	Shipbuilding	Subject	Pipe Volting

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Flange	· 5K 50A	PCS	2	For 1 person
	Gasket	· 5K 50A	PCS	1	For 1 person
	Hex Bolt	· M12, 45mm	PCS	4	For 1 person
	Hexagon Nut	· M12	PCS	4	For 1 person
	Washer	· M12	PCS	8	For 1 person
	Spanner	· 19mm	PCS	2	For 1 person
	Vernier Calipers	Digital Vernier CalipersReplace with a ruler if local supply is not available	PCS	1	For 1 person

Time Limit

6min

Cable binding

Watch the video

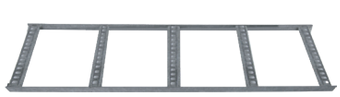
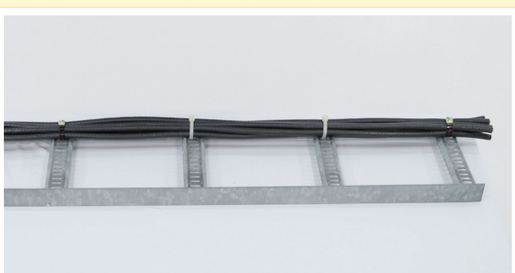
Job Category	Shipbuilding	Subject	Cable binding

Ⅰ Requirements

– Bind the cable with appropriate way and tools.

Ⅱ Diagram & Requirements

1) Diagram

CABLE TRAY	CABLE	After Binding
		

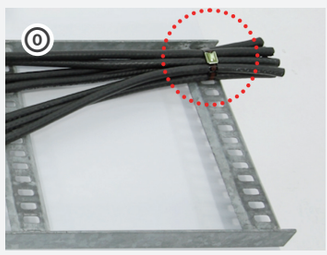
Job Category	Shipbuilding	Subject	Cable binding

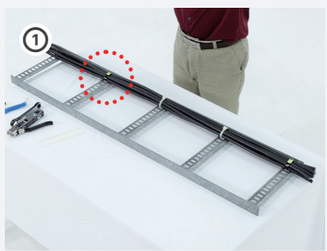
2) Requirements

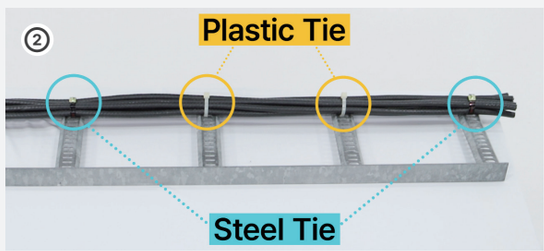
⓪ (Pre) Bind a bundle(7 strands) of cables with steel ties at the srarting point of the cable circuit.

① Bind the rest of the cable circuit so that it looks like the instuctor bound it as shown below.

② Bind both ends of the cable circuit with a steel tie and the middle with a plastic tie.







③ Put the plastic ties first from far away as shown in Figure Ⓐ, so that they can be pulled and tightened. And after binding, remove the ends that ar not related to cable fixing as shown in Figure Ⓑ.





Job Category	Shipbuilding	Subject	Cable binding

④ Bind the cable with a steel tie in the following order.







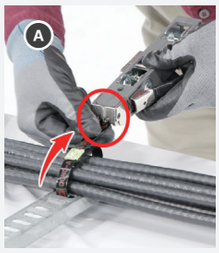
Ⓐ Shape the end of the steel tie into a semicircle for easy insertion into the cable circuit.

Ⓑ Put the steel tie first in a hole close to your body.(The opposite direction with plastic tie)

Ⓒ Insert the tip of the tie into the steel tie head.

Job Category	Shipbuilding	Subject	Cable binding

⑤ Bind the steel tie using a dedicated tool in the following order.











Ⓐ Insert the end of the steel tie into the entrance of the dedicated tool body.

ⒷⒸ Insert into the tool body so that the ratchet handle splits are inserted into the blade next to the steel tie.

Ⓓ Turn the ractchet handle and tighten the cable to prevent it from moving.

Ⓔ Tightened to the appropriate intensity, and remove the tip of the tie by grasping the cutting lever located on the left side of the body.

⑥ After the assesment, perform the dismantling operation to the pre-assessment state.

Job Category	Shipbuilding	Subject	Cable binding

Ⅲ Equipment & Materials

Equipment & Materials		Spec.	Unit	QTY	Remakrs
	Cable	· About (O/D)15 X (L)1,000	PCS	7	For 1 person
	Cable Binding tool	· 5K50A(Using a universal binder tool)	PCS	1	For 1 person
	Scissors	· For cutting plastic ties	PCS	1	For 1 person
	Steel Ties	· Length 11.8 inches, width 0.5 inches	PCS	1	For 1 person
	Plastic Ties	· Width 13mm, Length 300mm	PCS	2	For 1 person